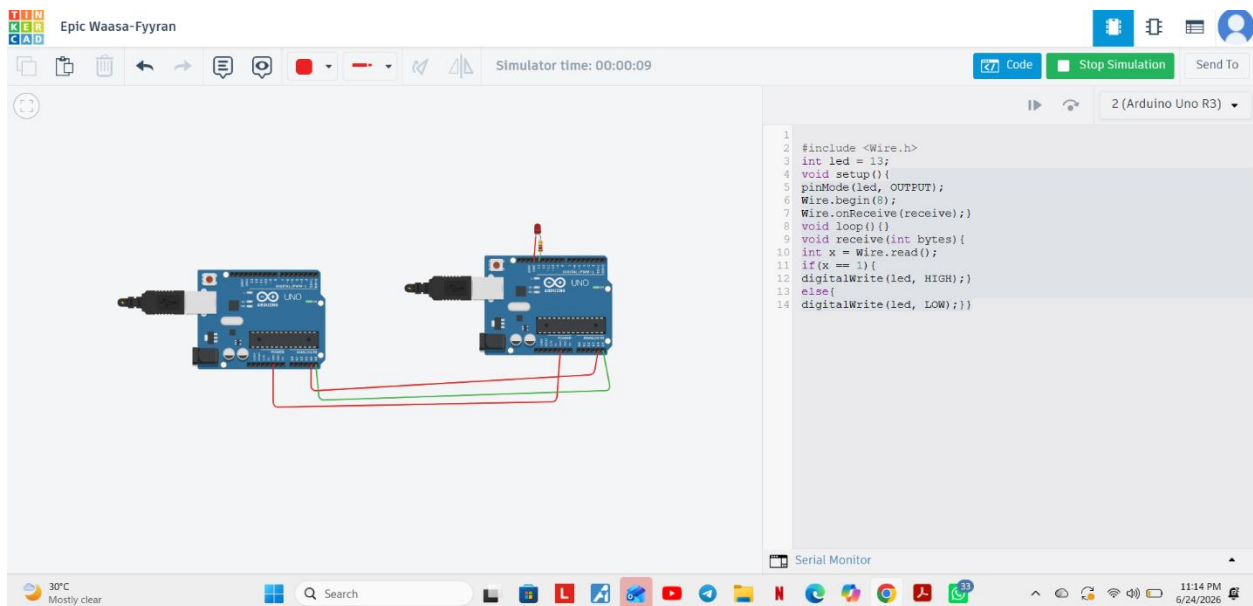


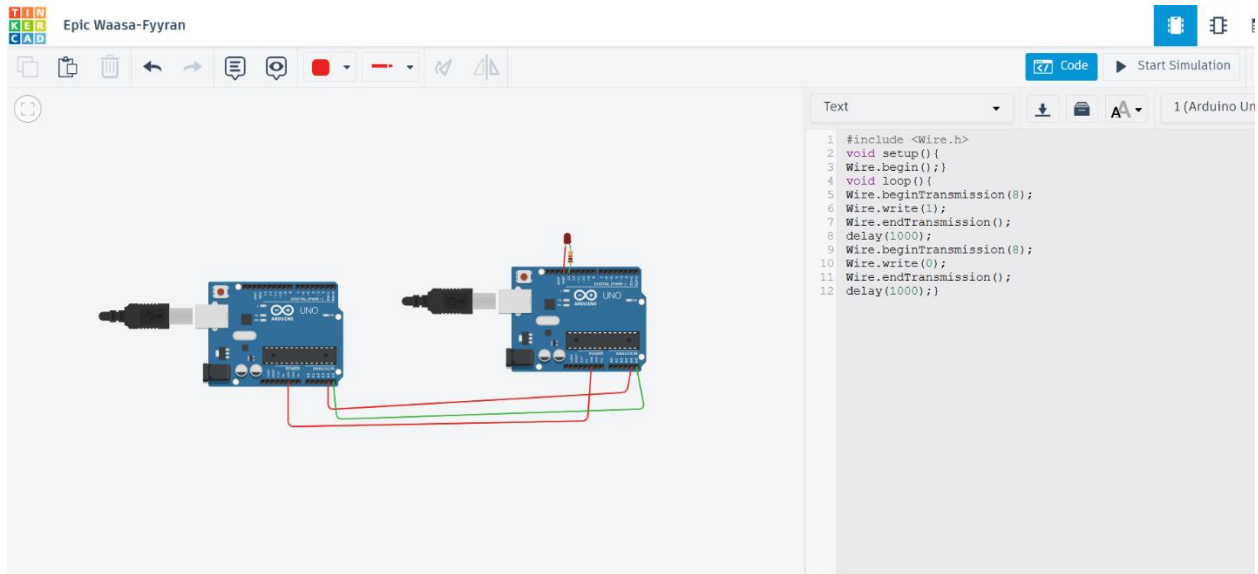
IOT ASSIGNMENT

1) How to enable sleep mode in ESP32?

On the ESP32, sleep mode is a way to conserve energy by reducing or shutting down parts of the chip when continuous operation isn't needed, which is especially useful in battery-powered or IoT devices. In light sleep, the CPU is paused but RAM and most peripherals remain powered, allowing for quick wakeups within milliseconds and a good balance between saving power and retaining data. Deep sleep, on the other hand, powers down the CPU, most RAM, and peripherals, leaving only the real-time clock and ultra-low-power co-processor active, which maximizes energy savings but requires longer wakeup times. To enable sleep, you first disable wireless peripherals like Wi-Fi and Bluetooth using functions such as `esp_wifi_stop()` and `esp_bt_controller_disable()`, then configure wakeup sources with APIs like `esp_sleep_enable_timer_wakeup()` for timed wakeups, `esp_sleep_enable_ext0_wakeup()` for GPIO triggers, or touchpad interrupts, and you can even combine multiple sources. Once everything is set, you call `esp_light_sleep_start()` for light sleep or `esp_deep_sleep_start()` for deep sleep, and when the ESP32 wakes up, you can check the reason using `esp_sleep_get_wakeup_cause()` to decide how your program should continue.

2) Setting up I2C in tinkercad.





3) Write a code to set up SPI.

```
#include <SPI.h>
```

```
void setup() {
```

```
    SPI.begin();
```

```
    SPI.beginTransaction(SPISettings(1000000, MSBFIRST, SPI_MODE0));
```

```
    Serial.begin(9600);
```

```
    Serial.println("SPI Master Initialized");
```

```
}
```

```
void loop() {
```

```
    digitalWrite(SS, LOW);
```

```
    byte response = SPI.transfer(0x42);
```

```
    digitalWrite(SS, HIGH);
```

```
    Serial.print("Received: ");
```

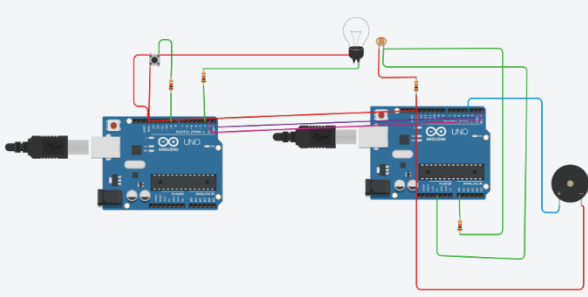
```
Serial.println(response);
```

```
delay(1000;
```

```
}
```

4) Establish Master Slave tx/rx between 2 arduino. Uart. Try transferring data.

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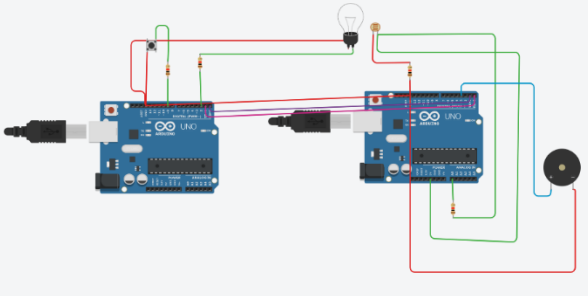
```
2 //
3 int led=2;
4 int button=9;
5 void setup()
6
7 {
8   pinMode(led, OUTPUT);
9   pinMode(button, INPUT_PULLUP);
10  Serial.begin(9600);
11 }
12
13 void loop()
14 {
15   if(digitalRead(button)==LOW) {
16     digitalWrite(led, HIGH);
17     delay(1000);
18   }
19   else{
20     digitalWrite(led, LOW);
21     delay(1000);
22   }
23 }
```

1 (Arduino Uno R3)

Serial Monitor

Send Clear

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```
1 // C++ code
2 //
3 int photoresist=A0;
4 int buzzer=3;
5 void setup()
6 {
7   pinMode(photoresist, INPUT_PULLUP);
8   pinMode(buzzer, OUTPUT);
9 }
10
11
12
13 void loop()
14 {
15   int brightness = digitalRead(photoresist);
16   Serial.println(brightness);
17   if(photoresist>=300){
18     buzzer==HIGH;
19   }
20   else{ buzzer==LOW;
21   }
22 }
```

2 (Arduino Uno R3)

Serial Monitor

Send Clear